

PROJECT ON CYBERSECURITY INCIDENT RESPONSE SIMULATION

1. Title of the Project

Cybersecurity Incident Response Simulation

2. Objectives

- To understand the cybersecurity incident response lifecycle.
- To investigate suspicious system and network activities.
- To collect and analyze digital evidence.
- To identify Indicators of Compromise (IoCs).
- To reconstruct the attack timeline.
- To recommend containment and recovery measures.
- To prepare a professional incident response report.

4. Tools Used

- Windows Event Viewer
- Process Explorer
- Task Manager
- TCP View
- Windows Security
- Wireshark

5. Environment Setup

- Windows 11
- VirtualBox Kali Linux

6. Investigation Methodology

Step-by-Step Procedure

Step 1: Environment Setup

- I started my Windows system and ensured an active internet connection.

- I opened the required tools: Process Explorer, Event Viewer, and Command Prompt.

Step 2: Process Analysis

- I opened Process Explorer and reviewed the running processes.
- I looked out for any unusual processes with high CPU or memory usage.

Step 3: Network Connection Analysis

- I ran netstat -ano in Command Prompt.
- I observed the active network connections, ports, and associated PIDs.

Step 4: Log Analysis

- I opened Windows Event Viewer.
- I reviewed the Security and System logs for failed logins, warnings, errors, and suspicious events.

Step 5: Indicator of Compromise (IoC) Identification

- I analyzed the collected evidence and identified any suspicious IP addresses, processes, login attempts, or network activity.

Step 6: Attack Timeline

- Opened command prompt and gave the command netstat -ano

Step 7: Wireshark Observation

- In kali Linux I opened wireshark and started capturing packets.
- Applied the filter and analyzed the TCP, DNS and ICMP packets.

Step 8: Attack Timeline

- I organized the observed events in chronological order to reconstruct the incident timeline.

7. Results

- Successfully monitored active system processes and analyzed their CPU and memory usage using Process Explorer.

- Examined Windows Security Event Logs to identify and interpret security-related events.
- Verified process details, including PID, company name, and parent-child relationships, to distinguish legitimate system processes.
- Observed different packets in Wireshark.
- No suspicious processes or abnormal system activities were detected during the analysis.
- Gained practical knowledge of Windows process monitoring and incident response techniques.

8. Conclusion

This project provided practical experience in monitoring system processes and analyzing Windows Security Event Logs using Process Explorer and Event Viewer. The analysis helped identify normal system activities, understand key security events, and reinforced the importance of these tools in detecting and investigating potential security incidents.

9. Screenshots

Records of Windows Security Logs through Event Viewer:

Date and Time: 29-06-2026 11:47:46

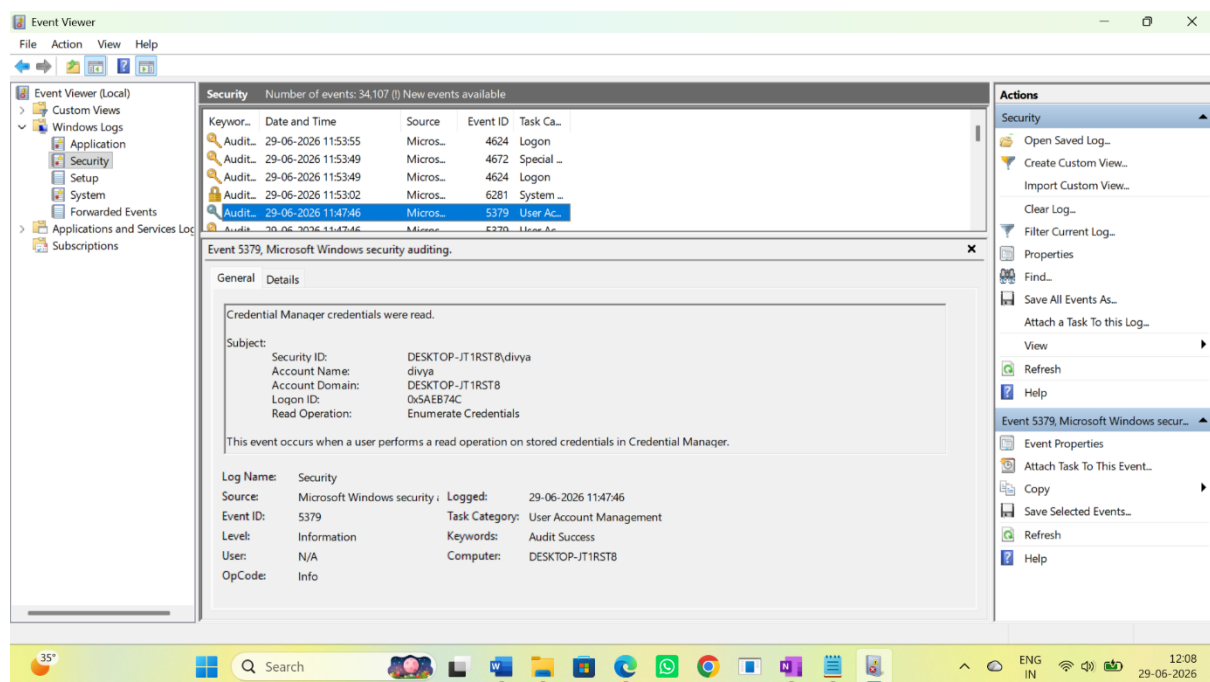
Event ID: 5379

Account Name: divya

Source: Microsoft Windows security auditing

Logon Type: 0

(Type 0 indicates that the event is associated with the Windows operating system itself, rather than a user logging in)



Date and Time: 27-06-2026 11:50:47

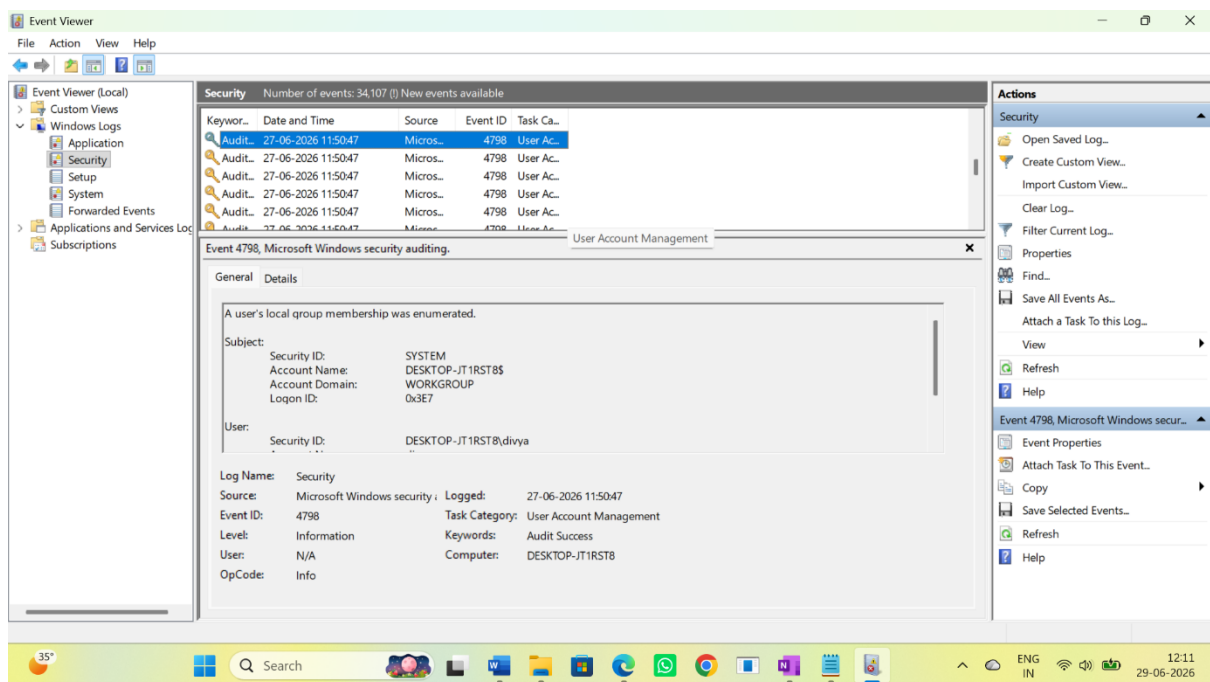
Event ID: 4798

Account Name: divya

Source: Microsoft Windows security auditing

Logon Type: -

(4798 records group membership enumeration, not a login therefore there is no Logon Type to record)



Date and Time: 29-06-2026 12:01:54

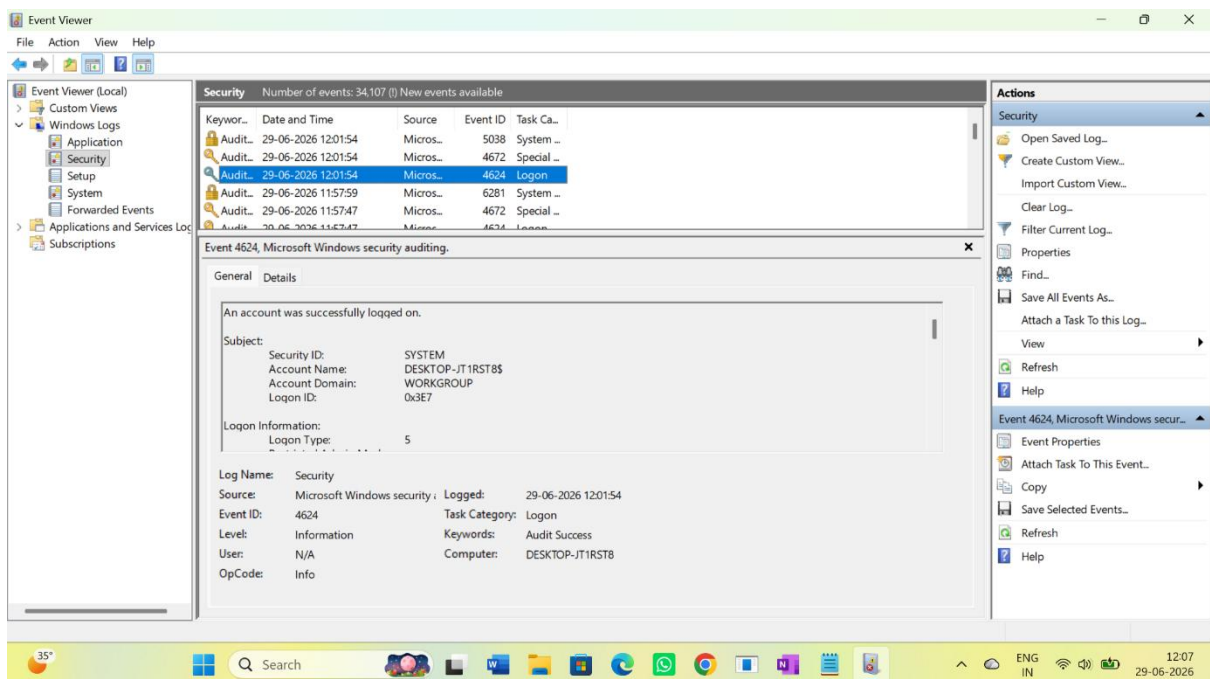
Event ID: 4624

Account Name: SYSTEM

Source: Microsoft Windows security auditing

Logon Type: 5

(Successful Login; Service Logon)



Recordings of Running Processes of the system through Process Explorer:

Process Name: svchost.exe

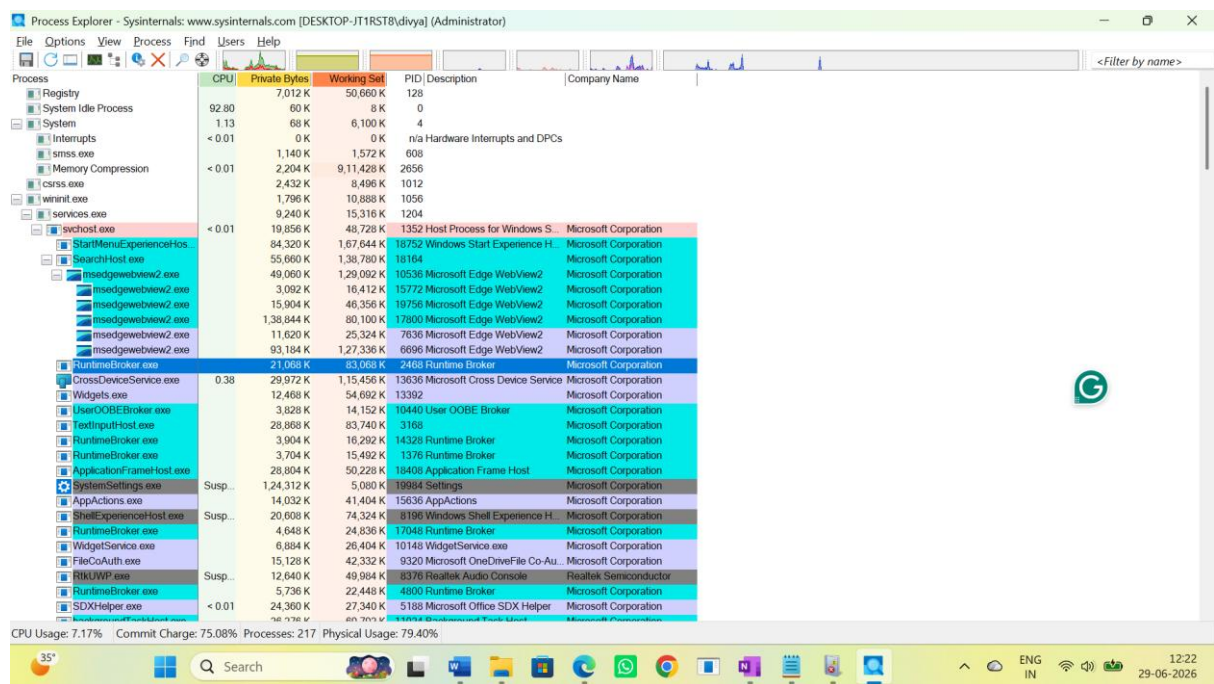
PID: 1352

Parent Process: services.exe(1204)

Company Name: Microsoft Corporation

CPU usage: <0.01

Memory usage: 48,564K



Process Name: cmdagent.exe

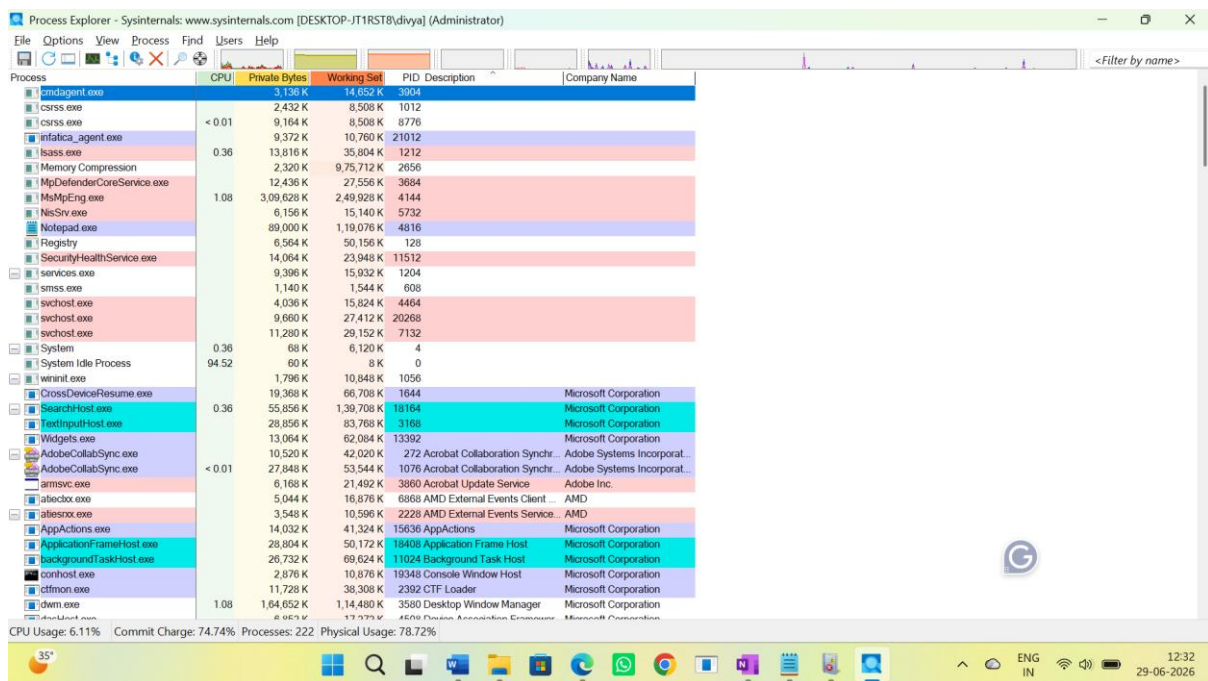
PID: 3904

Parent Process: services.exe(1204)

Company Name: Microsoft Corporation

CPU usage: -

Memory usage: 14,652K



Process	CPU	Private Bytes	Working Set	PID	Description	Company Name
cmdagent.exe		3,136 K	14,652 K	3904		
csrss.exe		2,432 K	8,508 K	1012		
csrss.exe	< 0.01	9,164 K	8,508 K	8776		
infatica_agent.exe		9,372 K	10,760 K	21012		
lsass.exe	0.36	13,816 K	35,804 K	1212		
Memory Compression		2,320 K	9,75,712 K	2656		
MpDefenderCoreService.exe		12,436 K	27,556 K	3684		
MsMpEng.exe	1.08	3,09,628 K	2,49,928 K	4144		
NsSvc.exe		6,156 K	15,140 K	5732		
Notepad.exe		89,000 K	1,19,076 K	4816		
Registry		6,564 K	50,156 K	128		
SecurityHealthService.exe		14,064 K	23,948 K	11512		
services.exe		9,396 K	15,932 K	1204		
smss.exe		1,140 K	1,544 K	608		
svchost.exe		4,036 K	15,824 K	4464		
svchost.exe		9,660 K	27,412 K	20268		
svchost.exe		11,280 K	29,152 K	7132		
System	0.36	68 K	6,120 K	4		
System Idle Process	94.52	60 K	8 K	0		
wininit.exe		1,796 K	10,848 K	1056		
CrossDeviceResume.exe		19,368 K	66,708 K	1644		Microsoft Corporation
SearchHost.exe	0.36	55,856 K	1,39,708 K	10164		Microsoft Corporation
TextinputHost.exe		28,856 K	83,768 K	3168		Microsoft Corporation
Widgets.exe		13,064 K	62,084 K	13392		Microsoft Corporation
AdobeCollabSync.exe		10,520 K	42,020 K	272	Adobe Acrobat Collaboration Synchr...	Adobe Systems Incorporat...
AdobeCollabSync.exe	< 0.01	27,848 K	53,544 K	1076	Adobe Acrobat Collaboration Synchr...	Adobe Systems Incorporat...
armexec.exe		6,168 K	21,492 K	3868	Adobe Acrobat Update Service	Adobe Inc.
alioexec.exe		5,044 K	16,876 K	6868	AMD External Events Client	AMD
alioexec.exe		3,548 K	10,596 K	2228	AMD External Events Service	AMD
AppActions.exe		14,032 K	41,324 K	15636	AppActions	Microsoft Corporation
ApplicationFrameHost.exe		28,804 K	50,172 K	18408	Application Frame Host	Microsoft Corporation
backgroundTaskHost.exe		26,732 K	69,624 K	11024	Background Task Host	Microsoft Corporation
conhost.exe		2,876 K	10,876 K	19348	Console Window Host	Microsoft Corporation
ctfmon.exe		11,728 K	38,308 K	2392	CTF Loader	Microsoft Corporation
dwm.exe	1.08	1,64,652 K	1,14,480 K	3580	Desktop Window Manager	Microsoft Corporation
dmoclient.exe		8,952 K	17,772 K	4600	Device Association Engine	Microsoft Corporation

CPU Usage: 6.11% Commit Charge: 74.74% Processes: 222 Physical Usage: 78.72%

35° ENG IN 12:32 29-06-2026

Records of TCP View:

[TCP] [System Process]

PID: 0

State: TIME_WAIT

Local: desktop-jt1rst8

Remote: server-18-161-229-112.maa50.r.cloudfront.net

[TCP] infatica_agent.exe

PID: 21012

State: ESTABLISHED

Local: desktop-jt1rst8

Remote: unn-89-222-107-247.datapacket.com

[TCP] infatica_agent.exe

PID: 21012

State: ESTABLISHED

Local: desktop-jt1rst8

Remote: unn-95-173-204-44.datapacket.com

[TCP] infatica_agent.exe

PID: 21012

State: ESTABLISHED

Local: desktop-jt1rst8

Remote: customer.clientshostname.com

[TCP] infatica_agent.exe

PID: 21012

State: ESTABLISHED

Local: desktop-jt1rst8

Remote: unn-152-233-46-133.datapacket.com

[TCP] infatica_agent.exe

PID: 21012

State: ESTABLISHED

Local: desktop-jt1rst8

Remote: customer.clientshostname.com

[TCP] infatica_agent.exe

PID: 21012

State: ESTABLISHED

Local: desktop-jt1rst8

Remote: .

[TCP] infatica_agent.exe

PID: 21012

State: ESTABLISHED

Local: desktop-jt1rst8

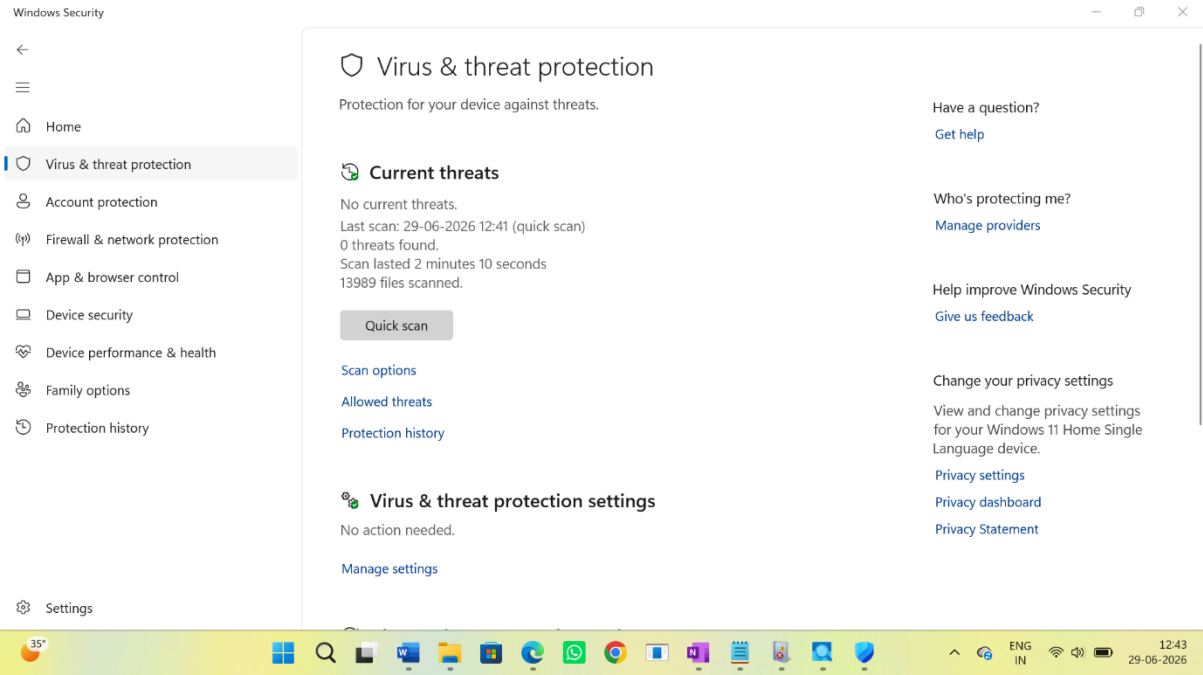
Remote: customer.clientshostname.com

```
C:\Users\divya\Downloads\Cyber law\TCPView\tcpvcon.exe
Tcpvcon.exe v4.19 - Sysinternals TcpVcon
Copyright (C) 1996-2023 Mark Russinovich & Bryce Cogswell
Sysinternals - www.sysinternals.com

[TCP] [System Process]
PID: 0
State: TIME_WAIT
Local: desktop-jt1rst8
Remote: server-18-161-229-112.maa50.r.cloudfront.net
[TCP] infatica_agent.exe
PID: 21012
State: ESTABLISHED
Local: desktop-jt1rst8
Remote: unn-89-222-107-247.datapacket.com
[TCP] infatica_agent.exe
PID: 21012
State: ESTABLISHED
Local: desktop-jt1rst8
Remote: unn-95-173-204-44.datapacket.com
[TCP] infatica_agent.exe
PID: 21012
State: ESTABLISHED
Local: desktop-jt1rst8
Remote: customer.clientshostname.com
[TCP] infatica_agent.exe
PID: 21012
State: ESTABLISHED
Local: desktop-jt1rst8
Remote: unn-152-233-46-133.datapacket.com
[TCP] infatica_agent.exe
PID: 21012
State: ESTABLISHED
Local: desktop-jt1rst8
Remote: customer.clientshostname.com
[TCP] infatica_agent.exe
PID: 21012
State: ESTABLISHED
Local: desktop-jt1rst8
Remote: .
```

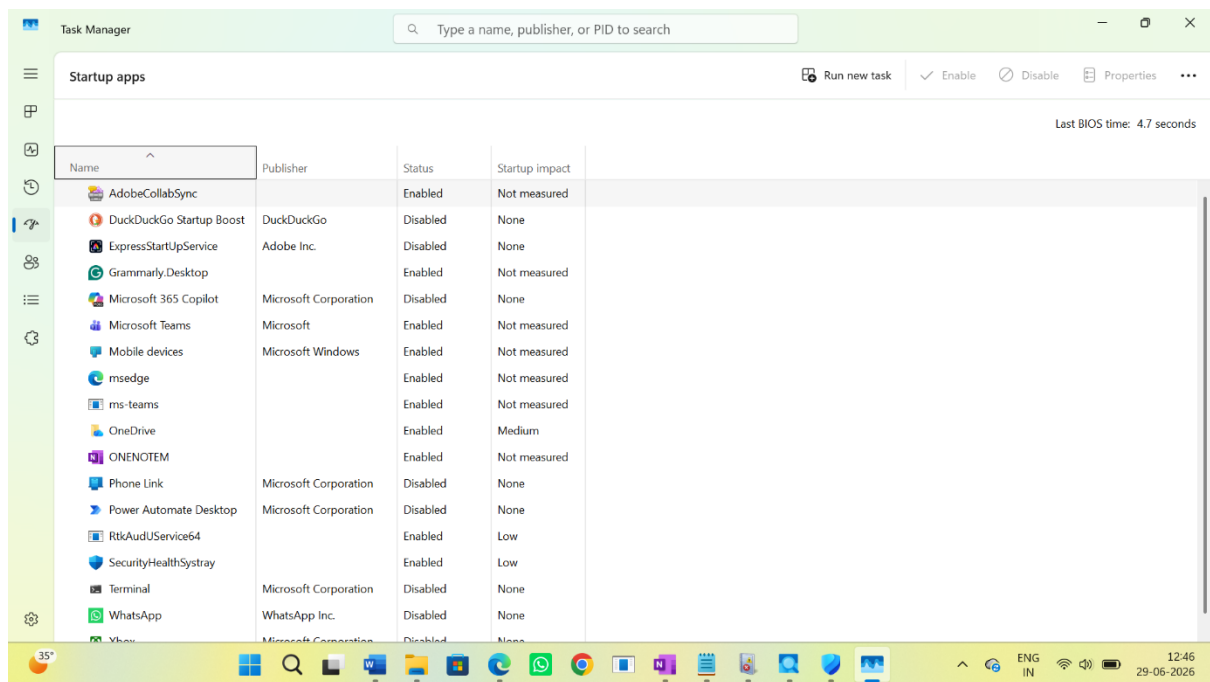
Records of Windows Defender:

No Threats Found.



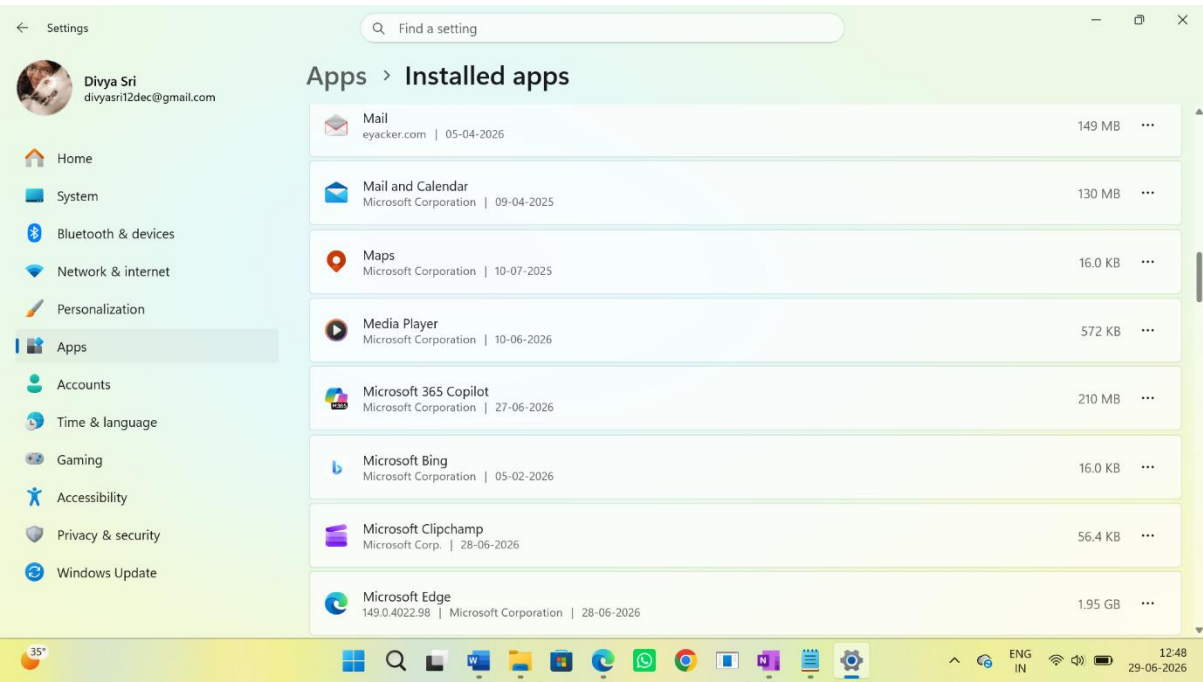
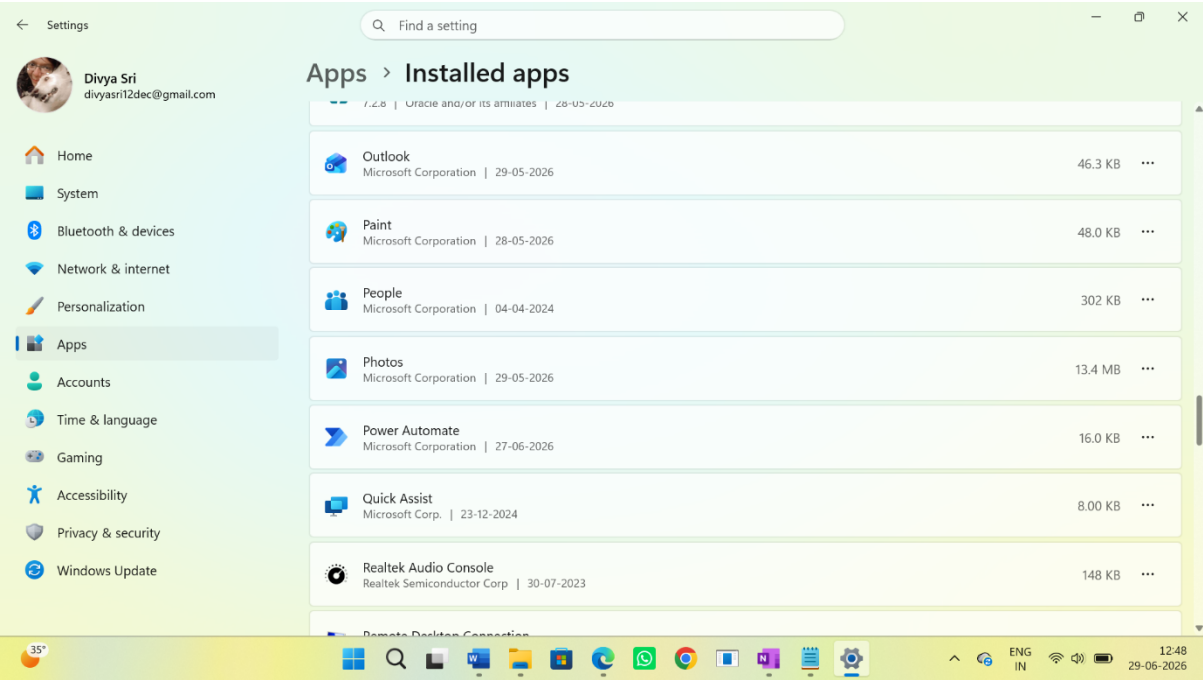
Records of Startup Apps:

No suspicious Startups were found.



Records of Installed Apps:

No suspicious apps were found.



Ran netstat -ano in command prompt:

Observed the ports that are Listening, Established, Closed and Time Wait.

```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.26200.8737]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>netstat -ano

Active Connections

Proto Local Address           Foreign Address         State       PID
TCP    0.0.0.0:135              0.0.0.0:0               LISTENING   1476
TCP    0.0.0.0:445              0.0.0.0:0               LISTENING    4
TCP    0.0.0.0:5040             0.0.0.0:0               LISTENING  6608
TCP    0.0.0.0:49664            0.0.0.0:0               LISTENING  1212
TCP    0.0.0.0:49665            0.0.0.0:0               LISTENING  1056
TCP    0.0.0.0:49666            0.0.0.0:0               LISTENING  1876
TCP    0.0.0.0:49667            0.0.0.0:0               LISTENING  2236
TCP    0.0.0.0:49668            0.0.0.0:0               LISTENING  3584
TCP    0.0.0.0:49669            0.0.0.0:0               LISTENING  1204
TCP    127.0.0.1:19294          0.0.0.0:0               LISTENING  1076
TCP    127.0.0.1:52007          127.0.0.1:65227         ESTABLISHED 18736
TCP    127.0.0.1:65227          0.0.0.0:0               LISTENING  10092
TCP    127.0.0.1:65227          127.0.0.1:52007         ESTABLISHED 18292
TCP    192.168.56.1:139         0.0.0.0:0               LISTENING    4
TCP    192.168.88.2:139         0.0.0.0:0               LISTENING    4
TCP    192.168.88.2:50132       89.222.107.247:8886     ESTABLISHED 21012
TCP    192.168.88.2:50145       95.173.204.44:8886     ESTABLISHED 21012
TCP    192.168.88.2:50148       185.162.130.27:8886    ESTABLISHED 21012
TCP    192.168.88.2:50149       152.233.46.133:8886    ESTABLISHED 21012
TCP    192.168.88.2:50150       185.162.130.219:8886   ESTABLISHED 21012
TCP    192.168.88.2:50151       162.244.34.11:8886     ESTABLISHED 21012
TCP    192.168.88.2:50152       185.162.130.21:8886    ESTABLISHED 21012
TCP    192.168.88.2:50154       185.162.130.221:8886   ESTABLISHED 21012
TCP    192.168.88.2:50155       162.244.34.17:8886     ESTABLISHED 21012
TCP    192.168.88.2:50156       185.162.130.8:8886     ESTABLISHED 21012
TCP    192.168.88.2:50157       185.162.130.26:8886    ESTABLISHED 21012
TCP    192.168.88.2:50158       185.162.130.28:8886    ESTABLISHED 21012
TCP    192.168.88.2:50159       162.244.34.4:8886      ESTABLISHED 21012
TCP    192.168.88.2:50160       162.244.34.42:8886     ESTABLISHED 21012
TCP    192.168.88.2:50161       185.162.128.91:8886    ESTABLISHED 21012
TCP    192.168.88.2:50162       178.249.213.82:8886    ESTABLISHED 21012
TCP    192.168.88.2:50163       185.223.94.98:8886     ESTABLISHED 21012
```

Records of Wireshark in Kali Linux:

The screenshot shows a Wireshark capture on the 'eth0' interface. The filter is set to 'dns'. The packet list shows a series of packets starting with a broadcast ARP request, followed by DNS queries and responses, and then a TLS handshake sequence. The packet details pane shows the structure of the selected packet, including Ethernet II, Internet Protocol, and Domain Name System fields. The packet bytes pane shows the raw data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	PCSSystemtec_8a:35:...	Broadcast	ARP	42	Who has 10.0.2.2? Tell 10.0.2.15
2	0.001122849	52:54:00:12:35:00	PCSSystemtec_8a:35:...	ARP	64	10.0.2.2 is at 52:54:00:12:35:00
3	0.001154124	10.0.2.15	192.168.88.1	DNS	95	Standard query 0x2022 A content-signature-2.cdn.mozilla.
4	0.001198226	10.0.2.15	192.168.88.1	DNS	95	Standard query 0xd72c AAAA content-signature-2.cdn.mozilla.
5	0.011528435	192.168.88.1	10.0.2.15	DNS	111	Standard query response 0x2022 A content-signature-2.cdn.
6	0.011529536	192.168.88.1	10.0.2.15	DNS	123	Standard query response 0xd72c AAAA content-signature-2.
7	0.017965299	10.0.2.15	34.160.144.191	TCP	74	48760 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_P
8	0.092625654	34.160.144.191	10.0.2.15	TCP	60	443 → 48760 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1
9	0.092733727	10.0.2.15	34.160.144.191	TCP	54	48760 → 443 [ACK] Seq=1 Ack=1 Win=64240 Len=0
10	0.093952129	10.0.2.15	34.160.144.191	TLSv1.2	274	Client Hello (SNI=content-signature-2.cdn.mozilla.net)
11	0.094761706	34.160.144.191	10.0.2.15	TCP	60	443 → 48760 [ACK] Seq=1 Ack=221 Win=65535 Len=0
12	0.192379615	34.160.144.191	10.0.2.15	TLSv1.2	1466	Server Hello
13	0.192504206	10.0.2.15	34.160.144.191	TCP	54	48760 → 443 [ACK] Seq=221 Ack=1413 Win=65535 Len=0
14	0.221545945	34.160.144.191	10.0.2.15	TLSv1.2	1466	Certificate
15	0.221669926	10.0.2.15	34.160.144.191	TCP	54	48760 → 443 [ACK] Seq=221 Ack=2825 Win=65535 Len=0
16	0.241850820	34.160.144.191	10.0.2.15	TLSv1.2	250	Server Key Exchange, Server Hello Done
17	0.241950370	10.0.2.15	34.160.144.191	TCP	54	48760 → 443 [ACK] Seq=221 Ack=3021 Win=65535 Len=0
18	0.257815119	10.0.2.15	34.160.144.191	TLSv1.2	147	Client Key Exchange, Change Cipher Spec, Encrypted Hands
19	0.258915098	34.160.144.191	10.0.2.15	TCP	60	443 → 48760 [ACK] Seq=3021 Ack=314 Win=65535 Len=0
20	0.396764457	34.160.144.191	10.0.2.15	TLSv1.2	365	New Session Ticket, Change Cipher Spec, Encrypted Handsh
21	0.400577654	10.0.2.15	34.160.144.191	TLSv1.2	153	Application Data
22	0.400844137	10.0.2.15	34.160.144.191	TLSv1.2	306	Application Data
23	0.401943455	34.160.144.191	10.0.2.15	TCP	60	443 → 48760 [ACK] Seq=3332 Ack=665 Win=65535 Len=0
24	0.406420274	34.160.144.191	10.0.2.15	TLSv1.2	123	Application Data
25	0.406943004	10.0.2.15	34.160.144.191	TLSv1.2	92	Application Data
26	0.409845944	34.160.144.191	10.0.2.15	TCP	60	443 → 48760 [ACK] Seq=3401 Ack=703 Win=65535 Len=0
27	0.452331308	34.160.144.191	10.0.2.15	TLSv1.2	1494	Application Data, Application Data
28	0.452332499	34.160.144.191	10.0.2.15	TLSv1.2	1494	Application Data
29	0.452333179	34.160.144.191	10.0.2.15	TLSv1.2	1494	Application Data
30	0.452333259	34.160.144.191	10.0.2.15	TLSv1.2	1494	Application Data

Frame 1: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336) on interface eth0, from PCSSystemtec_8a:35:d2 (08:00:27:8a:35:d2), Destination: Broadcast, Protocol: Address Resolution Protocol (request)

Domain Name System: Protocol

Packets: 28910 - Dropped: 0 (0.0%)

Profile: Default

The screenshot shows a Wireshark capture on the 'eth0' interface. The filter is set to 'dns'. The packet list shows a series of packets starting with a TCP connection, followed by a TLS handshake sequence, and then application data. The packet details pane shows the structure of the selected packet, including Ethernet II, Internet Protocol, and Domain Name System fields. The packet bytes pane shows the raw data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
55	1.471234452	34.160.144.191	10.0.2.15	TCP	60	443 → 48760 [ACK] Seq=14797 Ack=1046 Win=65535 Len=0
56	1.540928225	34.160.144.191	10.0.2.15	TLSv1.2	100	Application Data
57	1.540929226	34.160.144.191	10.0.2.15	TCP	60	443 → 48760 [FIN, ACK] Seq=14843 Ack=1046 Win=65535 Len=0
58	1.541021571	10.0.2.15	34.160.144.191	TCP	54	48760 → 443 [RST] Seq=1046 Win=0 Len=0
59	1.541087544	10.0.2.15	34.160.144.191	TCP	54	48760 → 443 [RST] Seq=1046 Win=0 Len=0
60	1.541639029	34.160.144.191	10.0.2.15	TCP	60	443 → 48760 [RST, ACK] Seq=4281591295 Ack=1046 Win=0 Len=0
61	2.696726071	10.0.2.15	192.168.88.1	DNS	97	Standard query 0x4756 A firefox.settings.services.mozilla.
62	2.696836626	10.0.2.15	192.168.88.1	DNS	97	Standard query 0x1851 AAAA firefox.settings.services.mozil
63	2.704401279	192.168.88.1	10.0.2.15	DNS	149	Standard query response 0x4756 A firefox.settings.servic
64	2.704402530	192.168.88.1	10.0.2.15	DNS	161	Standard query response 0x1851 AAAA firefox.settings.serv
65	2.705263128	10.0.2.15	151.101.157.91	TCP	74	42928 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_P
66	2.713959262	10.0.2.15	192.168.88.1	DNS	95	Standard query 0x5880 A content-signature-2.cdn.mozilla.
67	2.714715749	151.101.157.91	10.0.2.15	TCP	60	443 → 42928 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1
68	2.714826194	10.0.2.15	151.101.157.91	TCP	54	42928 → 443 [ACK] Seq=1 Ack=1 Win=64240 Len=0
69	2.717649178	10.0.2.15	192.168.88.1	DNS	95	Standard query 0x3889 AAAA content-signature-2.cdn.mozil
70	2.718676099	10.0.2.15	151.101.157.91	TLSv1.2	276	Client Hello (SNI=firefox.settings.services.mozilla.com)
71	2.719521659	151.101.157.91	10.0.2.15	TCP	60	443 → 42928 [ACK] Seq=1 Ack=223 Win=65535 Len=0
72	2.724576176	192.168.88.1	10.0.2.15	DNS	111	Standard query response 0x5880 A content-signature-2.cdn
73	2.724577327	192.168.88.1	10.0.2.15	DNS	123	Standard query response 0x3889 AAAA content-signature-2.
74	2.726978217	151.101.157.91	10.0.2.15	TLSv1.2	1494	Server Hello
75	2.726979147	151.101.157.91	10.0.2.15	TCP	1494	443 → 42928 [ACK] Seq=1441 Ack=223 Win=65535 Len=1440 [T
76	2.727075265	10.0.2.15	151.101.157.91	TCP	54	42928 → 443 [ACK] Seq=223 Ack=1441 Win=65535 Len=0
77	2.727111703	10.0.2.15	151.101.157.91	TCP	54	42928 → 443 [ACK] Seq=223 Ack=2881 Win=65535 Len=0
78	2.727523782	151.101.157.91	10.0.2.15	TLSv1.2	1494	Certificate
79	2.727524183	151.101.157.91	10.0.2.15	TLSv1.2	281	Server Key Exchange, Server Hello Done
80	2.727564833	10.0.2.15	151.101.157.91	TCP	54	42928 → 443 [ACK] Seq=223 Ack=4321 Win=65535 Len=0
81	2.727590266	10.0.2.15	151.101.157.91	TCP	54	42928 → 443 [ACK] Seq=223 Ack=4548 Win=65535 Len=0
82	2.732027062	10.0.2.15	151.101.157.91	TLSv1.2	147	Client Key Exchange, Change Cipher Spec, Encrypted Hands
83	2.733116254	151.101.157.91	10.0.2.15	TCP	60	443 → 42928 [ACK] Seq=4548 Ack=316 Win=65535 Len=0
84	2.739597033	151.101.157.91	10.0.2.15	TLSv1.2	312	New Session Ticket, Change Cipher Spec, Encrypted Handsh

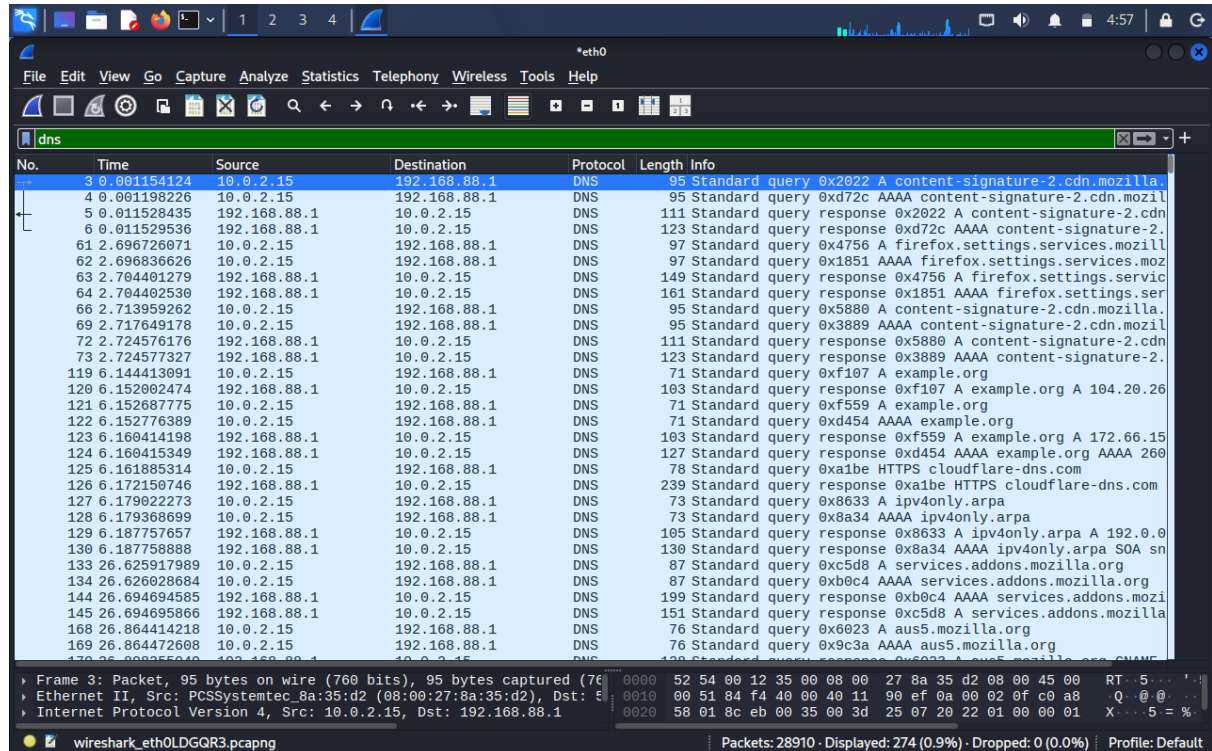
Frame 1: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336) on interface eth0, from PCSSystemtec_8a:35:d2 (08:00:27:8a:35:d2), Destination: Broadcast, Protocol: Address Resolution Protocol (request)

Domain Name System: Protocol

Packets: 28910 - Dropped: 0 (0.0%)

Profile: Default

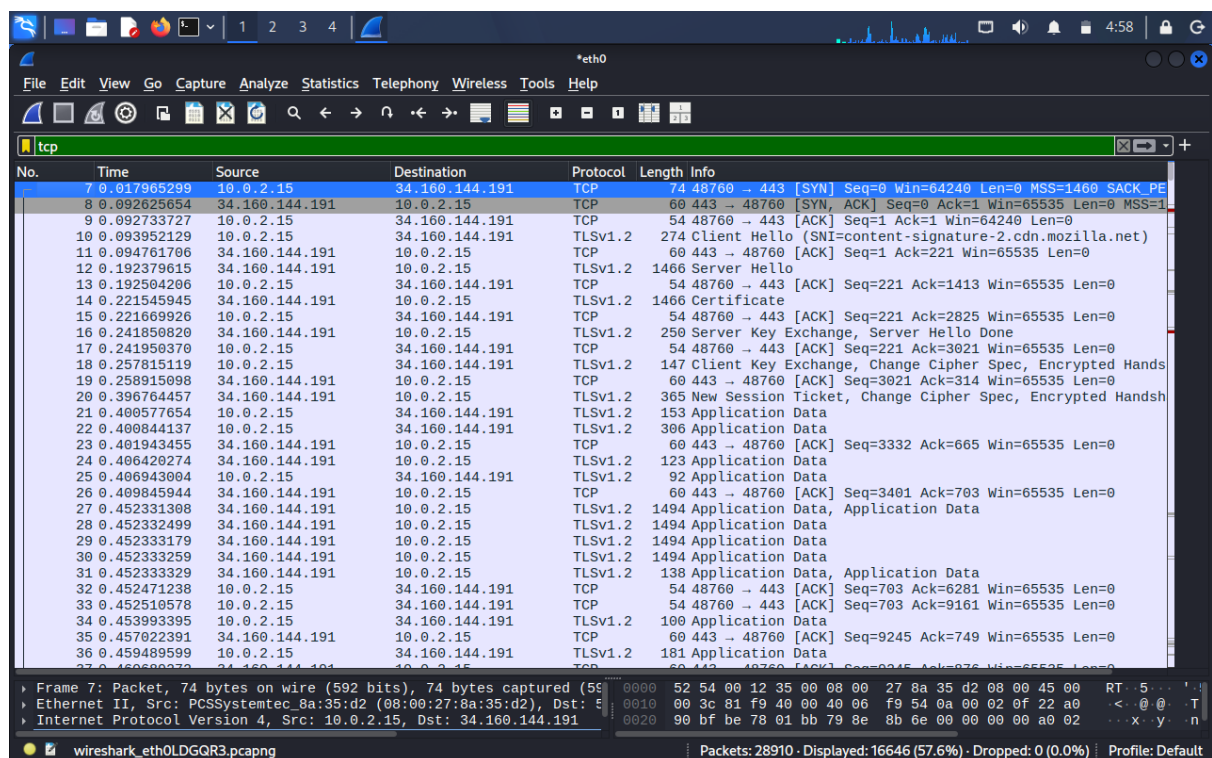
Filtered the packets through protocol DNS



The screenshot shows a Wireshark capture of network traffic on the eth0 interface, filtered by the protocol DNS. The packet list displays 95 packets, all of which are DNS queries or responses. The packet details pane shows the structure of a DNS packet, including the query type and the response data. The packet bytes pane shows the raw data of the selected packet.

No.	Time	Source	Destination	Protocol	Length	Info
3	0.001154124	10.0.2.15	192.168.88.1	DNS	95	Standard query 0x2022 A content-signature-2.cdn.mozilla.
4	0.001198226	10.0.2.15	192.168.88.1	DNS	95	Standard query 0xd72c AAAA content-signature-2.cdn.mozilla.
5	0.011528435	192.168.88.1	10.0.2.15	DNS	111	Standard query response 0x2022 A content-signature-2.cdn.mozilla.
6	0.011529536	192.168.88.1	10.0.2.15	DNS	123	Standard query response 0xd72c AAAA content-signature-2.cdn.mozilla.
61	2.696726071	10.0.2.15	192.168.88.1	DNS	97	Standard query 0x4756 A firefox.settings.services.mozilla.
62	2.696836626	10.0.2.15	192.168.88.1	DNS	97	Standard query 0x1851 AAAA firefox.settings.services.mozilla.
63	2.704401279	192.168.88.1	10.0.2.15	DNS	149	Standard query response 0x4756 A firefox.settings.services.mozilla.
64	2.704402530	192.168.88.1	10.0.2.15	DNS	161	Standard query response 0x1851 AAAA firefox.settings.services.mozilla.
66	2.713959262	10.0.2.15	192.168.88.1	DNS	95	Standard query 0x5880 A content-signature-2.cdn.mozilla.
69	2.717649178	10.0.2.15	192.168.88.1	DNS	95	Standard query 0x3889 AAAA content-signature-2.cdn.mozilla.
72	2.724576176	192.168.88.1	10.0.2.15	DNS	111	Standard query response 0x5880 A content-signature-2.cdn.mozilla.
73	2.724577327	192.168.88.1	10.0.2.15	DNS	123	Standard query response 0x3889 AAAA content-signature-2.cdn.mozilla.
119	6.144413091	10.0.2.15	192.168.88.1	DNS	71	Standard query 0xf107 A example.org
120	6.152092474	192.168.88.1	10.0.2.15	DNS	103	Standard query response 0xf107 A example.org A 104.20.26.104
121	6.152687775	10.0.2.15	192.168.88.1	DNS	71	Standard query 0xf559 A example.org
122	6.152776389	10.0.2.15	192.168.88.1	DNS	71	Standard query 0xd454 AAAA example.org
123	6.160414198	192.168.88.1	10.0.2.15	DNS	103	Standard query response 0xf559 A example.org A 172.66.15.172
124	6.160415349	192.168.88.1	10.0.2.15	DNS	127	Standard query response 0xd454 AAAA example.org AAAA 2601:1901:8080::1
125	6.161885314	10.0.2.15	192.168.88.1	DNS	78	Standard query 0xa1be HTTPS cloudflare-dns.com
126	6.172150746	192.168.88.1	10.0.2.15	DNS	239	Standard query response 0xa1be HTTPS cloudflare-dns.com
127	6.179022273	10.0.2.15	192.168.88.1	DNS	73	Standard query 0x8633 A ipv4only.arpa
128	6.179368699	10.0.2.15	192.168.88.1	DNS	73	Standard query 0x8a34 AAAA ipv4only.arpa
129	6.187757657	192.168.88.1	10.0.2.15	DNS	105	Standard query response 0x8633 A ipv4only.arpa A 192.0.2.192
130	6.187758888	192.168.88.1	10.0.2.15	DNS	130	Standard query response 0x8a34 AAAA ipv4only.arpa AAAA sn
133	26.625917989	10.0.2.15	192.168.88.1	DNS	87	Standard query 0xc5d8 A services.addons.mozilla.org
134	26.626028684	10.0.2.15	192.168.88.1	DNS	87	Standard query 0xb0c4 AAAA services.addons.mozilla.org
144	26.694694585	192.168.88.1	10.0.2.15	DNS	199	Standard query response 0xb0c4 AAAA services.addons.mozilla.org
145	26.694695866	192.168.88.1	10.0.2.15	DNS	151	Standard query response 0xc5d8 A services.addons.mozilla.org
168	26.864414218	10.0.2.15	192.168.88.1	DNS	76	Standard query 0x6023 A aus5.mozilla.org
169	26.864472608	10.0.2.15	192.168.88.1	DNS	76	Standard query 0x9c3a AAAA aus5.mozilla.org

Filtered the packets through protocol TCP



The screenshot shows a Wireshark capture of network traffic on the eth0 interface, filtered by the protocol TCP. The packet list displays 95 packets, all of which are TCP connections or data transfers. The packet details pane shows the structure of a TCP packet, including the sequence number, window size, and data length. The packet bytes pane shows the raw data of the selected packet.

No.	Time	Source	Destination	Protocol	Length	Info
7	0.017905299	10.0.2.15	34.160.144.191	TCP	74	48760 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PER
8	0.0922625654	34.160.144.191	10.0.2.15	TCP	60	443 → 48760 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1
9	0.092733727	10.0.2.15	34.160.144.191	TCP	54	48760 → 443 [ACK] Seq=1 Ack=1 Win=64240 Len=0
10	0.093952129	10.0.2.15	34.160.144.191	TLSv1.2	274	Client Hello (SNI=content-signature-2.cdn.mozilla.net)
11	0.094761706	34.160.144.191	10.0.2.15	TCP	60	443 → 48760 [ACK] Seq=1 Ack=221 Win=65535 Len=0
12	0.192379615	34.160.144.191	10.0.2.15	TLSv1.2	1466	Server Hello
13	0.192504206	10.0.2.15	34.160.144.191	TCP	54	48760 → 443 [ACK] Seq=221 Ack=1413 Win=65535 Len=0
14	0.221545945	34.160.144.191	10.0.2.15	TLSv1.2	1466	Certificate
15	0.221669926	10.0.2.15	34.160.144.191	TCP	54	48760 → 443 [ACK] Seq=221 Ack=2025 Win=65535 Len=0
16	0.241850820	34.160.144.191	10.0.2.15	TLSv1.2	250	Server Key Exchange, Server Hello Done
17	0.241950370	10.0.2.15	34.160.144.191	TCP	54	48760 → 443 [ACK] Seq=221 Ack=3021 Win=65535 Len=0
18	0.257815119	10.0.2.15	34.160.144.191	TLSv1.2	147	Client Key Exchange, Change Cipher Spec, Encrypted Hands
19	0.258915098	34.160.144.191	10.0.2.15	TCP	60	443 → 48760 [ACK] Seq=3021 Ack=314 Win=65535 Len=0
20	0.396764457	34.160.144.191	10.0.2.15	TLSv1.2	365	New Session Ticket, Change Cipher Spec, Encrypted Handsh
21	0.400577654	10.0.2.15	34.160.144.191	TLSv1.2	153	Application Data
22	0.400844137	10.0.2.15	34.160.144.191	TLSv1.2	306	Application Data
23	0.401943455	34.160.144.191	10.0.2.15	TCP	60	443 → 48760 [ACK] Seq=3332 Ack=665 Win=65535 Len=0
24	0.406420274	34.160.144.191	10.0.2.15	TLSv1.2	123	Application Data
25	0.406943004	10.0.2.15	34.160.144.191	TLSv1.2	92	Application Data
26	0.409845944	34.160.144.191	10.0.2.15	TCP	60	443 → 48760 [ACK] Seq=3401 Ack=703 Win=65535 Len=0
27	0.452331308	34.160.144.191	10.0.2.15	TLSv1.2	1494	Application Data, Application Data
28	0.452332499	34.160.144.191	10.0.2.15	TLSv1.2	1494	Application Data
29	0.452333179	34.160.144.191	10.0.2.15	TLSv1.2	1494	Application Data
30	0.452333259	34.160.144.191	10.0.2.15	TLSv1.2	1494	Application Data
31	0.452333329	34.160.144.191	10.0.2.15	TLSv1.2	138	Application Data, Application Data
32	0.452471238	10.0.2.15	34.160.144.191	TCP	54	48760 → 443 [ACK] Seq=703 Ack=6281 Win=65535 Len=0
33	0.452510578	10.0.2.15	34.160.144.191	TCP	54	48760 → 443 [ACK] Seq=703 Ack=9161 Win=65535 Len=0
34	0.453993395	10.0.2.15	34.160.144.191	TLSv1.2	100	Application Data
35	0.457822391	34.160.144.191	10.0.2.15	TCP	60	443 → 48760 [ACK] Seq=9245 Ack=749 Win=65535 Len=0
36	0.459489599	10.0.2.15	34.160.144.191	TLSv1.2	181	Application Data

10. Indicators of Compromise (IoCs)

During the investigation, the collected evidence from Windows Event Viewer, Process Explorer, TCPView, Windows Defender, Startup Applications, and Installed Applications was analyzed to identify potential Indicators of Compromise (IoCs).

The following observations were made:

IOC Category	Observation	Assessment
Windows Security Event	Event ID 5379	Normal operating system credential manager activity
Windows Security Event	Event ID 4798	Group membership enumeration; informational event
Windows Security Event	Event ID 4624 (Logon Type 5)	Successful service logon performed by the SYSTEM account
Running Processes	svchost.exe	Legitimate Microsoft Windows service process
Running Processes	cmdagent.exe	Security software process; no abnormal behaviour observed
Network Connections	Multiple established TCP connections	Active outbound connections from installed applications; no evidence of malicious communication identified during manual inspection
Windows Defender	No threats detected	System scan indicated no malware detection
Installed Applications	No suspicious software	All installed applications appeared legitimate

Based on the available evidence, no confirmed Indicators of Compromise were identified during the investigation.

11. Attack Timeline Reconstruction

The following timeline was reconstructed from the collected evidence:

Time	Event
29-06-2026 11:47:46	Event ID 5379 recorded. Windows Credential Manager credentials were accessed by the operating system.
27-06-2026 11:50:47	Event ID 4798 recorded. User account group membership was enumerated.
29-06-2026 12:01:54	Event ID 4624 recorded. Successful Service Logon (Logon Type 5) performed by the SYSTEM account.
29-06-2026 12:22:35	Running processes were examined using Process Explorer. No unauthorized or malicious processes were identified.
29-06-2026 12:37:27	Active network connections were reviewed using TCPView. Several established outbound connections from legitimate applications were observed.
29-06-2026 12:43:41	Windows Defender protection history was reviewed. No threats or malware detections were found.
29-06-2026 12:46:47	Startup applications and installed software were examined. No suspicious persistence mechanisms or unauthorized software were identified.
29-06-2026 13:17:28	Ran Command prompt netstat -ano

Incident Assessment

Although several Windows security events were observed, detailed analysis determined that they were consistent with normal operating system activity. No evidence of unauthorized access, malware execution, privilege escalation, or persistence mechanisms was identified.

Containment Actions

As no active compromise was confirmed, immediate containment was not required. However, the following preventive measures are recommended:

- Continue monitoring Windows Security Logs.
- Regularly review authentication events.
- Perform periodic antivirus scans.
- Keep Windows and installed applications updated.

- Enable multi-factor authentication where possible.
- Monitor unusual outbound network connections.
- Conduct periodic security audits.

12. Recovery Recommendations

No recovery actions were necessary because no system compromise was detected.

Preventive recommendations include:

- Maintain current security patches.
- Perform scheduled malware scans.
- Review user account privileges.
- Enable centralized log monitoring.
- Implement continuous network monitoring.
- Maintain regular system backups.

The investigation demonstrated the importance of correlating multiple evidence sources rather than relying on a single event. Individual Windows security events such as Event ID 5379, Event ID 4798, and Event ID 4624 may initially appear suspicious; however, when analyzed alongside process information, network activity, and endpoint security status, they were determined to represent legitimate operating system behaviour. This highlights the importance of validating security alerts before classifying them as incidents.

13. Github Link: <https://github.com/divyasri-web/Cybersecurity-Incident-Response-Simulation-.git>

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